

Generation Interconnection System Impact Study Report

For Queue Project Q19-1218

Kickapoo 100 kV

March 17, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1218. The Interconnection Customer proposes a 123.5 MW solar photovoltaic facility interconnecting at the Kickapoo 100 kV substation in the Ohio area. The request was received on June 03, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 339 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$30,466,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
HESPERIA - CHUB LAKE 100 KV	TREGO - SWAN VLY 500 KV	139.6	131	38.6
COVANTA HENNEPIN ENERGY - CHUB LAKE 100 KV	SWAN VLY - TREGO 500 KV	132.3	125	38.6
LAKESIDE 2 - COVANTA HENNEPIN ENERGY 100 KV	SWAN VLY - TREGO 500 KV	115.7	131	38.6
PLEASANT STREET - LAKE MARION 100 KV	HESPERIA - STRYKER 100 KV	113.6	112	31.7
DON - HATO REY GIS 1403-1404 100 KV L	POMARIA - TAIBAN MESA 100 KV	110.9	87	5.0
LAKESIDE 2 SUBSTATION - MERCED FALLS 100 KV Line	LANDSMAN CREEK - RUSSELLVILLE 100 KV	104.1	138	5.2
VLY CHILDREN'S HOSPITAL - PLEASANT STREET 100 KV	STRYKER - HESPERIA 100 KV	102.8	118	31.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HESPERIA - CHUB LAKE 100 KV	Rebuild line with higher-capacity conductor	\$5,714,000	100% (\$5,714,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
COVANTA HENNEPIN ENERGY - CHUB LAKE 100 KV	Rebuild line with higher-capacity conductor	\$4,711,000	100% (\$4,711,000)
LAKESIDE 2 - COVANTA HENNEPIN ENERGY 100 KV	Rebuild line with higher-capacity conductor	\$6,255,000	100% (\$6,255,000)
PLEASANT STREET - LAKE MARION 100 KV	Rebuild line with higher-capacity conductor	\$8,756,000	100% (\$8,756,000)
DON - HATO REY GIS 1403-1404 100 KV L	Rebuild line with higher-capacity conductor	\$1,757,000	100% (\$1,757,000)
MERCED FALLS - LAKESIDE 2 100 KV	Reconductor line	\$1,159,000	100% (\$1,159,000)
PLEASANT STREET - VALLEY CHILDREN'S HOSPITAL 100 KV line	Reconductor line	\$2,114,000	100% (\$2,114,000)
	Total allocated to project		\$30,466,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.